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I. INTRODUCTION

This technical report presents the thermo-mechanical design and structural verification of a Reactor Pressure Vessel and its internal components for a nuclear power plant. The primary objective is to ensure that the vessel maintains its structural integrity and safety margins under both normal and accidental operating conditions, in strict accordance with the standards established by the ASME Boiler and Pressure Vessel Code, Section III [1].

The study adopts a multi-physics methodology that integrates mechanical sizing with a detailed thermal analysis. The process begins with the classification of the vessel and the thermal shield into specific safety classes, quality groups, and seismic categories based on their functional importance and safety roles. Following this classification, the mechanical sizing of the vessel is performed by evaluating the thickness required to withstand internal pressure according to the Tresca criterion and by verifying the resistance to elastic instability under external pressure using the Corradi method.

A significant portion of the analysis is dedicated to the thermal behavior of the system, specifically modeling the volumetric heat generation caused by the absorption of γ radiation. This involves calculating the steady-state temperature profiles across the vessel wall and evaluating the necessity of a thermal shield. The shield is designed to mitigate secondary thermal stresses and prevent thermal ratcheting, ensuring that the entire assembly respects the elastic shakedown limit of $3S_m$. By following this iterative design approach, the report provides a validated configuration that satisfies both mechanical load requirements and thermal stability criteria, including considerations for material properties and neutron shielding.

It is important to emphasize that the analytical model adopted in this study is highly simplified and does not account for several critical physical phenomena. For instance, the analysis neglects the effects of corrosion, which in a real reactor environment would require the application of protective cladding layers; such layers would introduce additional thermal resistance and potentially degrade the overall heat exchange efficiency. Furthermore, the study relies on one-dimensional analytical solutions rather than a detailed CFD analysis, which would be necessary to capture complex three-dimensional flow patterns and local temperature variations. Finally, the long-term effects of radiation on material properties, such as neutron embrittlement and the resulting shift in the Nil Ductility Transition Temperature (NDT), are not considered in this preliminary assessment.

II. VESSEL

A. Classification

The purpose of this section is to identify and justify the classification (Quality Group, Safety Class, and Seismic Category) assigned to the reactor components and to define the criteria for the material selection. The analysis follows the guidelines of the *ASME Boiler and Pressure Vessel Code, Section III* [1] and the regulatory references provided [2].

The methodology involved reviewing the design functions,

identifying system boundaries, and applying classification criteria based on structural importance and safety margins.

Based on the analysis, the vessel is identified as a primary pressure boundary component, directly involved in containing the reactor coolant under normal and accidental conditions. Therefore, it is assigned the most stringent classifications:

- Safety Class 1 and Quality Group A: as it contains the primary coolant at high pressure, this ensures the highest level of fabrication control and inspection.
- Seismic Category I: to guarantee structural integrity during Design-Basis Earthquake (DBE) conditions.

Parameter	Reactor pressure vessel
Safety Class	1
Quality Group	A
Seismic Category	I

Tab. I: Summary of component classifications for the vessel.

B. Design conditions

The design conditions are established as conservative, fictitious parameters that provide the structural basis for sizing the component under ASME III Service Level A limits [1]. The internal design pressure (P_{des}) is defined by applying a safety margin to the operating pressure to account for potential transients (such as pressure surges, control system errors, or static head effects). Conversely, the external design pressure (P_{ext}) is explicitly set to match the operating pressure of the Containment for Primary system Protection, CPP. This choice represents the most severe physical scenario of primary system depressurization, where the vessel must withstand the full hydrostatic head of the surrounding still water. The Design Temperature (T_{des}) is specified to be no less than the maximum expected average metal temperature through the vessel thickness.

Internal P_{des} [bar]	External P_{ext} [bar]	T_{des} [$^{\circ}$ C]
82.5	75	334

Tab. II: Design conditions.

Since the final thickness is not known a priori but is determined through an iterative calculation, the design temperature T_{des} is recalculated at each iteration, taking as the first guess value the maximum outlet temperature equal to 270 $^{\circ}$ C. The table reports the final average temperature from the elastic collapse verification, as this represents the limiting failure criterion.

C. Thickness sizing

The determination of the reactor pressure vessel thickness follows a multi-physics approach, balancing mechanical integrity against thermal-induced stresses. To ensure structural safety under Service Level A limits, the component is evaluated against two bounding parameters: the internal design pressure, assuming a depressurized CPP during primary system operation, and the external design pressure, representing

a Loss of Coolant Accident (LOCA) scenario where the vessel must resist elastic instability, commonly referred to as buckling.

1) *Internal pressure*: For this section the Tresca criterion is adopted due to its inherent conservatism for ductile steel components. While Lamé solutions precisely describe radial stress variations, the Mariotte-based ASME III formula [1] is utilized to ensure that the onset of yielding at the inner radius is prevented, assuming a uniform stress distribution across the thickness t :

$$t = \frac{P_{des} R_{int}}{S_m - 0.5 P_{des}} \quad (1)$$

Given that material properties such as the allowable stress intensity (S_m) and the yield strength (S_y) are temperature-dependent, the sizing follows an iterative computational loop. This process begins with an initial thickness guess based on the maximum core outlet temperature, followed by a thermal coupling phase where the average wall temperature ($\bar{T}$) is evaluated by integrating the profile across the thickness. For each iteration, a new thickness and the corresponding average temperature are calculated, for each iteration the stress intensity S_m is updated, if needed. The process continues until the average temperature $\bar{T}$ reaches convergence within a tolerance of 1 °C. The final stable values for both thickness and temperature are summarized in Table III.

Final thickness [cm]	Avg. Temp. [°C]	Tolerance [°C]
10.87	314	0

Tab. III: Final thickness results for the vessel using the Tresca criterion.

2) *External pressure*: The verification under external pressure requires a distinct analysis. In this regime, the transition between plastic collapse and elastic buckling is defined by the critical slenderness, which depends on the Young's modulus (E), the Poisson coefficient (ν), and the temperature-dependent yield strength (S_y):

$$\left(\frac{D}{t}\right)_{cr} = \sqrt{\frac{E}{S_y(1-\nu^2)}} \quad (2)$$

The Corradi design procedure is adopted [3], in which the following safety condition is verified through a dual-loop iteration:

$$p_{ext} \leq p_a = \frac{p_c}{S_{corradi}} \quad (3)$$

The safety margin $S_{corradi}$ is chosen equal to 2 because it makes the design comparable, on the average, to that produced by ASME, where sufficient numerical and experimental evidence exists to ensure that the code considers the proper safety margin.

Throughout the iterations, the ovality W is maintained within ASME III prescribed tolerances, and the thickness is incrementally increased by 0.1 mm until the allowable pressure (p_a) satisfies the external load requirement.

Final Thickness [cm]	Average temp. [°C]	Number of iterations
16.15	334	528

Tab. IV: Final thickness results for the vessel using the Corradi criterion.

The slenderness is then calculated, with the new thickness of 16.15 cm, and compared with $(D/t)_{cr}$ to ensure the vessel is within the plastic-dominant region.

The definitive vessel thickness is ultimately selected by adopting the most restrictive value between the analyzed load cases. In this project, the requirement to prevent elastic instability under external pressure proved to be the limiting factor.

D. Heat transfer coefficients

The convective heat transfer coefficient at the inner vessel surface (h_1), interacting with the primary coolant, was calculated using the Dittus-Boelter correlation for turbulent flow:

$$Nu = 0.023 \cdot Re^{0.8} Pr^{0.4} = \frac{h_1 D_e}{k_I} \quad (4)$$

Where D_e represents the hydraulic diameter of the annulus between the vessel and the core barrel, defined as $D_e = D_{ves} - D_{bar}$ and k_I is the primary fluid thermal conductivity. The convective coefficient at the outer surface (h_2), describing the natural convection from the thermal insulation to the containment water pool, was evaluated using the McAdams correlation instead. A reference temperature difference of $\Delta T = 30^\circ C$ was assumed for the calculation of the Grashof number:

$$Nu = 0.13 \cdot (Gr Pr)^{1/3} = \frac{h_2 L_{ext}}{k_{II}} \quad (5)$$

Where L_{ext} is the height of the vessel and k_{II} is the containment water thermal conductivity.

The global heat transfer coefficient at the outer surface, U_2 , accounts for the series thermal resistance of the insulation layer (conduction) and the external boundary layer (convection). It is defined relative to the external surface area:

$$U_2 = \left[\frac{R_{ext,ins}}{k_{ins}} \ln \left(\frac{R_{ext,ins}}{R_{ves} + t_{ves}} \right) + \frac{1}{h_2} \right]^{-1} \quad (6)$$

Where k_{ins} is the insulator's thermal conductivity and $R_{ext,ins} = R_{ves} + t_{ves} + t_{ins}$, being R_{ves} the internal radius of the vessel, t_{ves} the thickness of the vessel wall and t_{ins} the thickness of the thermal insulator.

For the subsequent thermal analysis, the global heat transfer coefficient at the surface between the vessel and the thermal insulator, U_1 , was also evaluated. Due to the cylindrical geometry, the heat transfer coefficients vary with radius; however, the heat transfer capacity remains conserved such that the product of the coefficient and its respective heat exchange area is constant:

$$U_1 \cdot A_1 = U_2 \cdot A_2 \quad (7)$$

The calculated values for both surfaces are summarized in Table V.

Parameter	Value
Inner convective coefficient h_1 [W/m^2K]	7498
Outer convective coefficient h_2 [W/m^2K]	1060.5
Outer global coefficient U_2 [W/m^2K]	26.90

Tab. V: Convective and global heat transfer coefficients.

E. Volumetric heat source

The determination of the volumetric heat source, q''' , within the reactor vessel wall induced by γ radiation absorption is critical for the subsequent thermal stress analysis as it represents the source term in the Fourier equation. The interaction of γ radiation with matter is modeled using the Lambert-Beer law for intensity attenuation, which is corrected by a build-up factor to account for scattering effects:

$$I(r) = I_0 B \cdot e^{-\mu r} \quad (8)$$

Where $I(r)$ is the radiation intensity at a depth r , B is the build-up factor, and μ is the linear attenuation coefficient of the steel. The incident intensity I_0 corresponds to the energy flux at the inner surface, defined as the product of the γ flux (Φ) and the average γ energy of the flux (E_γ):

$$I_0 = q_0'' = \Phi E_\gamma \quad (9)$$

Physically, the volumetric heat generation $q'''(r)$ represents the energy deposited per unit volume per unit time, which corresponds to the spatial rate of energy loss of the radiation beam. Therefore, it is derived by differentiating the intensity profile with respect to the radial coordinate r :

$$q'''(r) = -\frac{dI}{dr} \quad (10)$$

Substituting the Lambert-Beer expression into this derivative, the governing equation for the heat source profile becomes:

$$q'''(r) = \mu(\Phi E_\gamma)B \cdot e^{-\mu r} \quad (11)$$

This analytical model was implemented to generate the radial profile of the volumetric heat source across the vessel thickness, as illustrated in Figure 1. The calculated volumetric heat source at the vessel's inner and outer surfaces are summarized in Table VI.

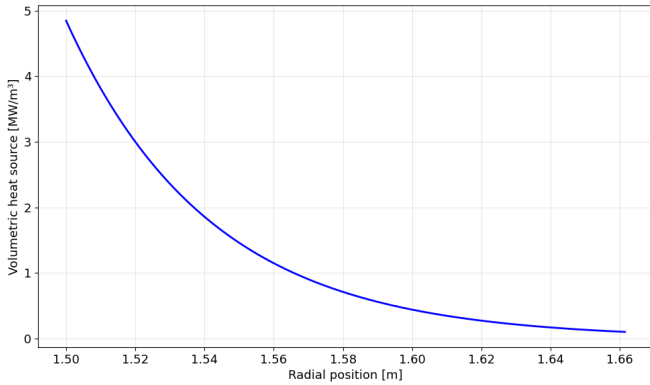


Fig. 1: Radial profile of the volumetric heat source across the vessel.

Boundary	Volumetric heat [MW/m^3]
Inner vessel surface	4.84
Outer vessel surface	0.10

Tab. VI: Volumetric heat source at the boundaries of the vessel.

F. Stationary temperature profile

The steady-state temperature profile across the vessel thickness is determined by solving the one-dimensional Fourier heat conduction equation, considering also the exponentially decaying volumetric heat source, $q'''(x)$:

$$\frac{d^2T}{dx^2} = -\frac{q_0'''}{k_s} e^{-\mu x} \quad (12)$$

Where k_s is the thermal conductivity of the steel. The general solution takes the form:

$$T(x) = -\frac{q_0'''}{\mu^2 k_s} e^{-\mu x} + Ax + B \quad (13)$$

The integration constants A and B are determined by applying boundary conditions at the vessel surfaces. At the inner surface ($x = 0$), the heat flux matches the convective transfer with the primary coolant, while at the outer surface ($x = t$), the flux accounts for conduction through the insulation and natural convection to the containment environment:

$$-k_s \left. \frac{dT}{dx} \right|_{x=0} = h_1(T_{\infty_1} - T(0)) \quad (14)$$

$$-k_s \left. \frac{dT}{dx} \right|_{x=t} = \frac{T(t) - T_{\infty_2}}{\frac{t_{ins}}{k_{ins}} + \frac{1}{h_2}} \quad (15)$$

Here T_{∞_1} is the primary fluid bulk temperature and T_{∞_2} is the containment water bulk temperature. The resulting temperature distribution is illustrated in Figure 2. Due to the internal heat generation, the peak temperature occurs within the wall thickness rather than at the boundaries. The critical thermal values are summarized in Table VII.

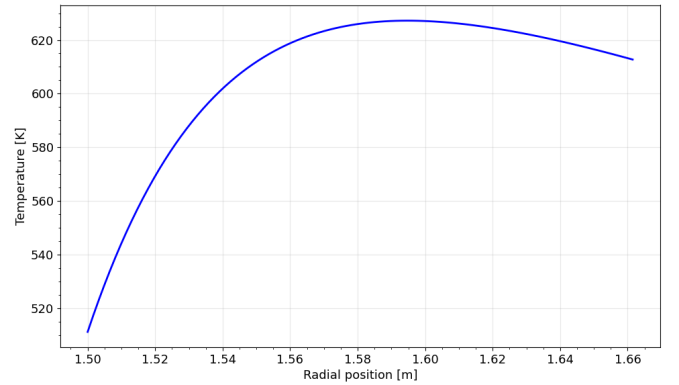


Fig. 2: Temperature profile across the vessel in stationary conditions.

Parameter	Value
Inner surface temperature	511 °K (238 °C)
Outer surface temperature	613 °K (339 °C)
Maximum temperature	627 °K (354 °C)
Position of Max. temperature	9.46 cm

Tab. VII: Key thermal parameters of the vessel wall in stationary conditions.

G. Comparative thermal analysis

To quantify the impact of γ radiation on the vessel's thermal regime, the calculated temperature profile was compared against the temperature profile obtained neglecting internal heat generation.

As illustrated in Figure 3, the absence of the volumetric source results in a standard monotonic temperature decay characteristic of pure radial conduction. On the other hand, the absorption of γ radiation induces a significant thermal bump, elevating the peak temperature within the wall well above the fluid interaction temperatures. This non-linear distribution is the primary driver of the secondary thermal stresses (Q) analyzed in subsection II-H.

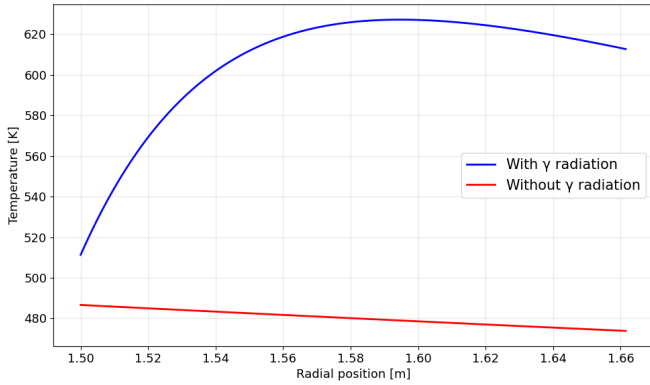


Fig. 3: Comparison of temperature profiles across the vessel: with γ heating (blue) vs. pure conduction (red).

The corresponding heat fluxes passing through the inner and outer surfaces of the vessel in the absence of the heat source were also evaluated and are summarized in Table VIII.

Boundary	Heat flux [kW/m^2]
Inner surface with primary coolant	4.01
Outer surface with containment	3.62

Tab. VIII: Thermal power flux at the boundaries of the vessel.

H. Resistance verification

The structural assessment of the reactor pressure vessel is conducted in accordance with the ASME Boiler and Pressure Vessel Code, Section III, Division 1 [1], following the "Design by Analysis" approach. Under Service Level A conditions, the methodology requires a rigorous classification of stresses to ensure high reliability through the evaluation of their nature and effect on the cylindrical structure.

To simplify the analytical model the vessel is modeled as an infinitely long, thin-walled cylinder remote from structural discontinuities, allowing for the assumption of a plane strain condition ($\epsilon_z = 0$). The material is treated as isotropic with linear elastic behavior, a hypothesis consistent with the objective of maintaining the structural response within the elastic shakedown limit.

Within this framework, stresses are categorized to distinguish between primary loads, necessary for mechanical equilibrium, and secondary loads, which are strain-controlled and self-limiting. Primary general membrane stress (P_m) arises from the internal design pressure and is evaluated to prevent immediate plastic collapse. In the undisturbed shell section considered here, primary local membrane (P_L) and primary bending (P_b) components are neglected. Secondary stresses (Q) are developed to satisfy geometric and thermal constraints; specifically, in the absence of a thermal shield, the vessel is subjected to the full radial temperature gradient induced by unattenuated γ radiation which induces extreme thermal stresses.

The assessment adopts the Guest-Tresca maximum shear stress theory as the governing failure criterion, providing a robust safety margin compared to the Von Mises theory. Structural integrity is verified by comparing the calculated stress intensities against the limits defined in NB-3200 [1].

First, to prevent plastic collapse under mechanical pressure, the general membrane stress must satisfy the limit:

$$P_m \leq S_m \quad (16)$$

Utilizing the ASME formula [1]:

$$P_m = P_{des} \frac{R_{ves}}{t} + \frac{P_{des}}{2} \quad (17)$$

the calculated membrane stress is 80.764 MPa, which remains well below the allowable limit of 114 MPa. This confirms that the initial vessel sizing is adequate for sustaining the primary mechanical loads.

However, the verification for Service Level A also requires the prevention of thermal ratcheting, or incremental collapse, by ensuring that the combined primary and secondary stress intensities satisfy the shakedown limit:

$$P_L + P_b + P_m + Q = S_{total} \leq 3S_m \quad (18)$$

The primary mechanical stresses, determined through Lamé's analytical solutions, are combined with the maximum thermal stresses, evaluated using the relation:

$$Q = \sigma_T \cdot \frac{\alpha E q''' }{k(1-\nu)\mu^2} \quad (19)$$

With a dimensionless factor $\sigma_T = 0.56$, derived from Figure 2 in [4] considering the reference geometry and the product μa .

The total stresses reach 510.18 MPa, significantly exceeding the shakedown limit of 342 MPa. The violation of the criterion implies that the vessel fails to achieve elastic shakedown; instead, the material enters a regime of thermal ratcheting where cyclic thermal gradients induce incremental plastic deformation.

To restore compliance and ensure long-term structural integrity, the secondary stress Q must be mitigated by reducing the volumetric heat source q''' . Consequently, the implementation of a thermal shield is necessary to intercept γ radiation and lower the thermal gradient within the vessel wall to values compatible with the shutdown threshold.

III. THERMAL SHIELD

A. Classification

Selecting the correct heat shield classification is critical in order to meet nuclear safety design requirements. Unlike the vessel, the shield is not part of the pressure boundary, leading to the following classification:

- Safety Class 2: the heat shield is not part of the primary system pressure barrier (a function performed by the vessel), so it is not a Class 1 component. However, it is a critical internal component, whose function is to ensure the structural integrity of the vessel by protecting it from excessive thermal stress. Structural failure could result in the generation of loose parts, potentially causing impact damage to the vessel or obstructing coolant flow. Given its importance in protecting a Class 1 component, it is classified as Safety Class 2.
- Quality Group B: the Quality Group defines the level of "Quality Assurance" (QA). In standard regulations, there is a direct correlation between Safety Class and Quality Group. Consistent with the assignment to Safety Class 2, Quality Group B is selected.
- Seismic Category I: being positioned coaxially between the barrel and the vessel, the heat shield is part of the reactor's internals. It must maintain its structural integrity and position during the Safe Shutdown Earthquake (SSE) to avoid impacting critical components (core, vessel) and to avoid altering the geometry of the coolant flow.

Parameter	Thermal shield
Safety Class	2
Quality Group	B
Seismic Category	I

Tab. IX: Summary of component classifications for the thermal shield.

B. Material selection

The selection of the thermal shield material is driven by its primary function of intercepting γ radiation, thereby lowering the heat flux arriving at the vessel's surface. Consequently, the material requires a high linear attenuation coefficient (μ) and high density to maximize photon absorption. In the energy range typical of reactor γ spectra (comprising prompt fission gammas and capture gammas), the dominant interaction mechanisms are the photoelectric effect at low energies, Compton scattering in the intermediate range, and pair production above 1.02 MeV.

For the photoelectric effect, the microscopic cross-section scales approximately as:

$$\sigma_{ph} \approx 5 \cdot 10^{11} \frac{Z^5}{E_\gamma^{3.5}} \quad (20)$$

which indicates a strong dependence on the atomic number Z . Compton scattering, on the other hand, is essentially material-independent at the microscopic level and enters through the macroscopic cross-section $\Sigma_{cp} = \sigma_{cp}ZN$, while pair production scales roughly with Z^2 for the energies of interest.

Given these interaction laws, high- Z and high-density materials provide more efficient attenuation of the uncollided beam. However, since Compton scattering dominates in the intermediate energy region most relevant for reactor shielding, the mass attenuation coefficient μ/ρ becomes relatively similar among different elements. In this regime, shielding effectiveness depends primarily on material density and only secondarily on Z . The choice of a high- Z material therefore provides a limited but non-negligible advantage, especially due to the more rapid absorption of secondary radiation.

Operatively, as the shield is immersed in the primary fluid at high temperatures, it demands excellent corrosion resistance and adequate mechanical strength (S_m , S_y) under service conditions. To manage internal heat generation effectively, the material must possess high thermal conductivity (k) to dissipate heat and a low coefficient of thermal expansion (α) to minimize secondary thermal stresses (Q). Finally, a low neutron activation cross-section is preferred to reduce induced radioactivity.

To satisfy these requirements and ensure full thermo-mechanical compatibility the same steel selected for the vessel has been adopted for the thermal shield.

C. Design conditions

Since the thermal shield is fully submerged within the primary coolant, the fluid pressures on its inner and outer surfaces are balanced, resulting in a negligible net differential pressure. However, to ensure high reliability and structural robustness against potential hydrodynamic transients or accidental flow conditions, the vessel's design pressure of 82.5 bar has also been adopted for this component.

As already said in subsection II-B, the design temperature (T_{des}) shall not be lower than the maximum average temperature expected under Level A service conditions, this parameter is evaluated considering the temperature profile carried out in subsection III-E.

Internal P_{des} [bar]	External P_{ext} [bar]	Design temp. T_{des} [°C]
82.5	82.5	222

Tab. X: Design conditions.

According to ASME III [1], the allowable stress must satisfy limits for both primary and secondary stresses. Detailed verification against the membrane limit ($P_m \leq S_m$) and the secondary stress range ($P + Q \leq 3S_m$) is performed at subsection III-F.

D. Thickness sizing

The thickness s of the thermal shield is calculated as the thickness necessary to ensure that thermal stresses in the vessel wall drop sufficiently to satisfy the shakedown criterion.

$$\sigma_{Lame} + Q \leq 3S_m \quad (21)$$

Where σ_{Lame} is the maximum mechanical stress intensity calculated via Lamé's equations and Q is the secondary thermal stress.

Substituting the analytical expression for Q , we solve for the maximum allowable volumetric heat generation $q'''_{allowable}$ at the vessel inner wall:

$$q'''_{allowable} = \frac{k(1-\nu)\mu^2}{\sigma_T \cdot \alpha E} (3S_m - \sigma_{Lame}) \quad (22)$$

Once the maximum allowable heat generation is known, we relate it to the initial unshielded source q'''_0 using the Lambert-Beer attenuation law:

$$q'''_{allowable} = q'''_0 \cdot e^{-\mu s} \quad (23)$$

Finally, inverting this relationship yields the required thickness:

$$s = -\frac{1}{\mu} \ln \left(\frac{q'''_{allowable}}{q'''_0} \right) = 2.13 \text{ cm} \quad (24)$$

This thickness ensures that the γ flux reaching the vessel is sufficiently attenuated, keeping the total stress state within the elastic shakedown limit.

The above calculation is performed by neglecting the attenuation of γ rays in water.

E. Stationary temperature profile

To determine the radial temperature profile, a steady-state thermal analysis is performed assuming a slab geometry, as prescribed.

The volumetric heat generation rate due to γ radiation is given by:

$$q'''(x) = q'''_0 e^{-\mu x} \quad (25)$$

The temperature distribution $T(x)$ is governed by the 1D heat conduction equation:

$$k \frac{d^2 T(x)}{dx^2} + q'''_0 e^{-\mu x} = 0 \quad (26)$$

The boundary conditions (Cauchy type) which account for convection on both surfaces are:

$$-k_s \left. \frac{dT}{dx} \right|_{x=0} = h_1 (T_b - T(0)) \quad (27)$$

$$-k_s \left. \frac{dT}{dx} \right|_{x=s} = h_1 (T(s) - T_b) \quad (28)$$

By solving the differential equation with the specified boundary conditions, the following values were obtained:

Parameter	Value
Inner surface temperature ($x = 0$)	493 °K (220 °C)
Outer surface temperature ($x = t$)	492 °K (219 °C)
Maximum temperature (T_{max})	497 °K (224 °C)
Position of maximum temperature (x_{max})	0.99 cm

Tab. XI: Key thermal parameters of the shield in stationary conditions.

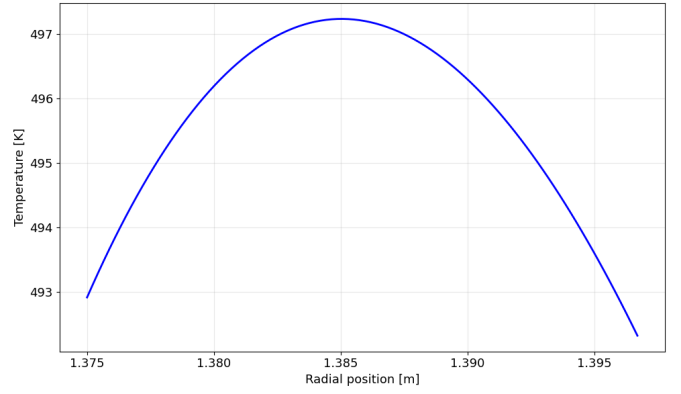


Fig. 4: Temperature profile inside the thermal shield.

F. Resistance verification

The structural assessment of the thermal shield follows the theoretical framework and stress categorization previously established for the reactor vessel. However, the unique operational role of the shield necessitates a distinct set of assumptions regarding its loading conditions.

Unlike the pressure vessel, the thermal shield is fully submerged in the primary coolant, which results in balanced internal and external pressures ($\Delta P \approx 0$). Consequently, primary membrane stresses (P_m) are considered negligible.

Under these conditions, the structural integrity is primarily dictated by secondary stresses (Q), which arise under a steep radial temperature gradient. The core of the verification process focuses on the shakedown limit, defined by the condition

$$Q \leq 3S_m \quad (29)$$

The secondary stress intensity Q is determined by evaluating the thermal stress components across the shield and subsequently applying the Tresca criterion to identify the maximum shear stress.

$$Q = 14.45 \text{ MPa} \ll 3S_m = 390 \text{ MPa} \quad (30)$$

These results confirm that the thermal shield operates under highly conservative conditions. The significant gap between the calculated secondary stress and the allowable limit guarantees that the component will achieve immediate elastic shakedown, ensuring long-term dimensional stability and structural reliability throughout its service life.

G. Positioning analysis

It is essential to evaluate the qualitative effects of the choice of the thermal shield inner radius, $R_{in,s}$, as this parameter significantly influences the thermal-hydraulic and structural performance of the system:

- Thermal-hydraulic impact: the introduction of the thermal shield divides the downcomer into two distinct flow regions. Since mass flow rate is conserved and fluid density is assumed constant, the reduction in the available cross-sectional area leads to an increase in the primary fluid velocity.

Due to the cylindrical geometry, the position of a shield

with a fixed thickness significantly affects the magnitude of this velocity increase. Specifically:

- Positioning the shield closer to the barrel (smaller radius) results in a lower area reduction, with a calculated velocity increase of approximately 6%.
- Positioning the shield closer to the vessel (larger radius) leads to a higher area occupancy, resulting in a velocity increase of approximately 7.5%.
- Convective heat transfer coefficient: the convective heat exchange coefficient (h_1) is governed by the Dittus-Boelter correlation:

$$Nu = 0.023 \cdot Re^{0.8} Pr^{0.4} \quad (31)$$

By expanding the terms it can be observed that h_1 is inversely proportional to the hydraulic diameter (D_h) according to $h \propto v^{0.8} D_h^{-0.2}$.

Consequently, reducing the gap between the shield and the vessel (i.e. moving the shield outward) decreases the local hydraulic diameter, which enhances the convective cooling efficiency on the internal surfaces. This suggests that placing the shield closer to the vessel is preferable from a thermo-hydraulic standpoint, as it optimizes heat removal while minimizing the overall flow blockage compared to an outer placement.

- Radiation attenuation and shielding: while the thickness is the primary factor for γ attenuation, the inner radius determines the geometric view factor for the radiation flux impacting the shield. Furthermore, the volume of water between the shield and the vessel acts as an additional layer of shielding; moving the shield closer to the vessel reduces the amount of water in between, potentially increasing the neutron and γ flux reaching the vessel wall. It should also be noted that while the calculations in the previous points were performed assuming a simplified slab geometry with a constant incident flux, a real reactor configuration involves a cylindrical geometry. In this case, the radiation flux undergoes not only exponential attenuation due to material interaction but also a reduction due to geometric divergence. As a result, the flux is expected to decrease (approximately by 20% between the barrel and the vessel). This implies that positioning the thermal shield too close to the barrel would expose the shield itself to a significantly higher radiation field. This would lead to a higher volumetric heat generation rate (q'''), potentially resulting in excessive thermal stresses and steeper temperature gradients within the shield, which might exceed the allowable limits for Service Level A.
- Thermal boundary layer and shield placement: in this analysis, the flow is assumed to be fully developed turbulent flow. A strategic consideration for the radial placement of the thermal shield is its position relative to the thermal boundary layer generated by the barrel. Considering a reference reactor length of 7 meter, calculations suggest that the shield should be positioned at a distance of at least 23 cm from the barrel. This ensures that the shield is placed beyond the barrel's thermal boundary layer, allowing the coolant to

reach the shield at the nominal core inlet temperature (214 °C). This prevents the shield from being cooled by "pre-heated" fluid, which would otherwise reduce the temperature gradient at the wall and impair heat removal. Furthermore, this optimal spacing is essential to avoid the overheating of the pressure vessel; by ensuring efficient heat removal from the shield and maintaining a proper water gap, the thermal stresses on the vessel are kept within the safety limits, preventing excessive temperature peaks on its inner surface.

- Economic and material considerations: from an engineering and manufacturing perspective, the radial position of the shield has a direct impact on material costs. Since the thermal shield is a cylindrical component, the volume of material required for a fixed thickness and height is a function of its mean radius. By positioning the shield closer to the barrel, thus reducing the mean radius, the total volume of steel is minimized compared to a placement closer to the vessel. This reduction in material not only lowers the procurement costs but also reduces the total weight of the reactor internals, potentially simplifying the support structures and the assembly procedures during construction.

IV. CONSIDERATIONS ON NEUTRON SHIELD

While the thermal shield is primarily designed to attenuate γ radiation and reduce thermal stresses, a comprehensive design must also address the fast neutron flux to prevent radiation damage and thermal stresses in the reactor pressure vessel. For this application, borated stainless steel is selected as the primary material. This choice is justified by its multi-functional properties:

- Neutron absorption: the presence of ^{10}B , which has an extremely high thermal neutron capture cross-section, allows for efficient removal of neutrons after they have been moderated by the surrounding primary water.
- Structural integrity: it maintains the mechanical strength and corrosion resistance typical of the austenitic steels used in PWR internals, ensuring compatibility with the primary coolant environment.

The neutron shield is ideally placed between the barrel and the thermal shield for the following reasons:

- Spectral softening: by placing the neutron shield closer to the core, the neutron spectrum is "softened" (shifted to lower energies) before reaching the thermal shield and the vessel.
- Secondary γ mitigation: neutron capture reactions (such as in Boron or Steel) often produce secondary γ rays. Placing the neutron shield inside the thermal shield allows the latter to attenuate these secondary γ rays before they reach the vessel wall.

V. VESSEL AND THERMAL SHIELD

For the purpose of the coupled vessel and thermal shield analysis, the shield was assumed to be positioned exactly in the middle of the annular region between the barrel and the vessel, $D_{shield} = 2.75 \text{ m}$. This approximation simplifies the

calculation of the geometric attenuation and the definition of the water gaps on both sides. While in a real design the position is optimized for thermal-hydraulic reasons (as discussed in subsection III-G), this centered placement allows for a balanced evaluation of the primary and secondary coolant volumes, providing a representative estimate of the overall radiation shielding effectiveness.

A. Design conditions

The introduction of the thermal shield modifies the thermal environment of the vessel while leaving the mechanical pressure requirements unchanged. The design temperature must be re-evaluated through a new iterative cycle. By reducing the volumetric heat generation within the vessel wall, the maximum average metal temperature through the thickness is significantly lowered compared to the unshielded case.

Internal P_{des} [bar]	External P_{ext} [bar]	T_{des} [°C]
82.5	75	272

Tab. XII: Design conditions for the shielded vessel.

B. Thickness sizing

This reduction in volumetric heat generation allows for a lower average wall temperature $\bar{T}$, which directly impacts the material's allowable stress S_m and the resulting structural dimensions.

1) *Internal pressure*: The sizing process was updated by repeating the iterative procedure for internal pressure, accounting for the reduced radiation source at the vessel's inner radius. As shown in Table XIII, the attenuated thermal gradient leads to a lower design temperature and a final thickness.

Final thickness [cm]	Avg. Temp. [°C]	Tolerance [°C]
9.11	259	0

Tab. XIII: Final thickness results for the shielded vessel using the Tresca criterion.

2) *External pressure*: The same procedure as before is carried out taking as starting thickness the final value obtained from the internal pressure sizing. Ultimately, elastic instability under external load remains the limiting design factor as shown in Table XIV.

Final Thickness [cm]	Average temp. [°C]	Number of iterations
15.22	272	611

Tab. XIV: Final thickness results for the shielded vessel using the Corradi criterion.

The required thickness against external pressure decreased from 16.15 cm to 15.22 cm due to a drop in the design temperature from 334 °C to 272 °C. At lower temperatures the yield strength of the steel is higher, providing greater resistance to buckling.

C. Heat transfer coefficients

The methodology for calculating the convective heat transfer coefficients remains consistent with the one adopted in subsection II-D. The only difference lies in the evaluation of h_1 , where the values for the primary fluid velocity and the hydraulic diameter are updated to account for the specific geometry of the considered region.

Parameter	Value
Inner convective coefficient h_1 [W/m^2K]	9273
Outer convective coefficient h_2 [W/m^2K]	1060.5
Outer global coefficient U_2 [W/m^2K]	26.89

Tab. XV: Convective and global heat transfer coefficients.

D. Volumetric heat source

The behavior of the volumetric heat source is identical to that of the unshielded case, the sole difference lies in the value of q_0''' .

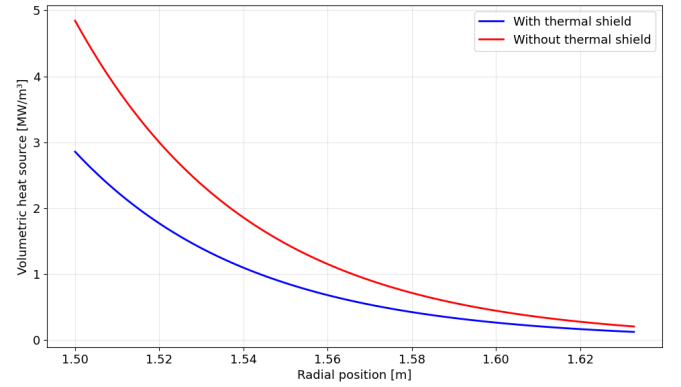


Fig. 5: Radial profile of the volumetric heat source across the shielded vessel.

Boundary	Volumetric heat [MW/m^3]
Inner vessel surface	2.86
Outer vessel surface	0.07

Tab. XVI: Volumetric heat source at the boundaries of the shielded vessel.

E. Stationary temperature profile

The thermal profile is substantially modified by the presence of the thermal shield, which minimizes the incident heat flux. Consequently, both the peak internal temperature and the surface temperatures remain closer to the primary coolant temperature, significantly reducing the magnitude of the thermal stresses developed within the vessel.

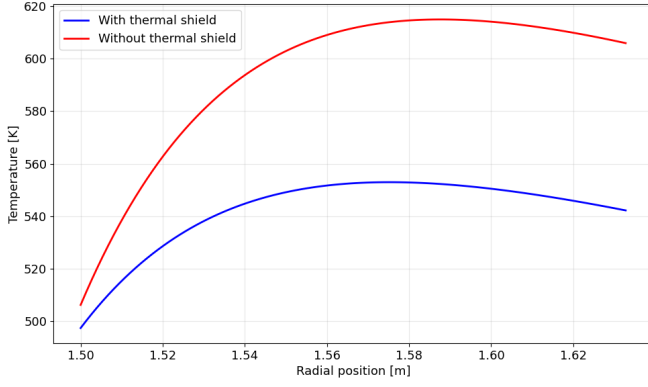


Fig. 6: Temperature profile across the shielded vessel in stationary conditions.

A significant advantage of the shielded configuration is the mitigation of creep-related risks. The theoretical threshold for the onset of creep is typically defined as $0.3T_m$, where T_m is the melting temperature. For steel alloys employed in nuclear applications ($T_m \approx 1500^\circ\text{C} = 1773^\circ\text{K}$), which corresponds to approximately 259°C .

In the unshielded configuration, the maximum wall temperature reached 354°C , well exceeding the temperature limit for creep. The thermal shield limits the maximum temperature to 283°C , while this value technically exceeds the strict threshold, it represents a marginal deviation compared to the unshielded case and remains well below the temperatures required to induce significant time-dependent deformation.

Parameter	Value
Inner vessel temperature ($x = 0$)	498°K (225°C)
Outer vessel temperature ($x = t$)	542°K (269°C)
Maximum temperature (T_{max})	556°K (283°C)
Position of maximum temperature (x_{max})	7.99 cm

Tab. XVII: Key thermal parameters of the shielded vessel in stationary conditions.

F. Comparative thermal analysis

The profile remains the same since the heat generation coming from the γ flux is not considered. The difference in the values of the inner and outer thermal power can be explained with the different thickness of the vessel compared to before. The results are reported at Table XVIII and the two temperature profiles are shown at Figure 7.

Boundary	Heat flux [kW/m^2]
Inner surface with primary coolant	4.01
Outer surface with containment	3.64

Tab. XVIII: Thermal power flux at the boundaries of the shielded vessel.

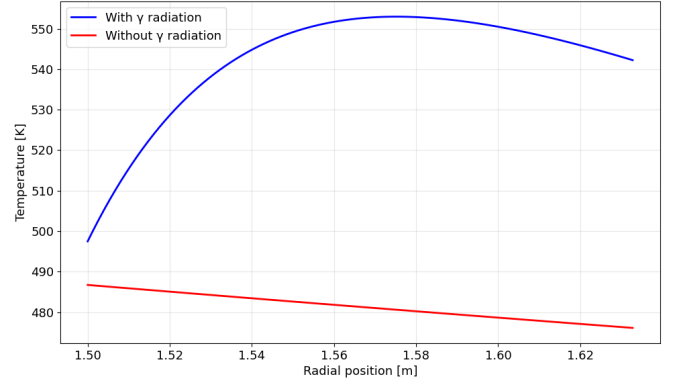


Fig. 7: Comparison of temperature profiles across the shielded vessel: with γ heating (blue) vs. pure conduction (red).

G. Resistance verification

While the primary mechanical loads remain constant, the secondary stress profile is significantly altered by the presence of the internal shielding.

The safety margin of the shielded vessel is evaluated using the combined stress intensity:

$$(P_m + Q)_{shielded} \leq 3S_m \quad (32)$$

The primary mechanical stresses, determined through Lamé's analytical solutions, are combined with the secondary thermal stresses; the principal stress components are obtained by superimposing the respective mechanical and thermal contributions. The total stress intensity is then derived by applying the Tresca criterion, calculating the maximum difference between the combined principal stresses, and subsequently comparing them against the allowable limit to ensure compliance with the elastic shakedown requirement.

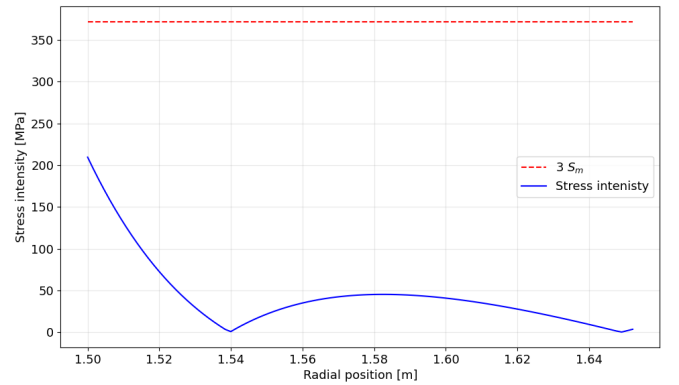


Fig. 8: Comparison of the maximum stress intensity profile against the allowable limit.

The updated calculation, incorporating the thermal shield, yields a maximum combined stress intensity of $\sigma_{tot} = 209.58\text{ MPa}$, which is compared against the allowable limit below.

$$\sigma_{tot} = 209.58\text{ MPa} < 3S_m = 372\text{ MPa} \quad (33)$$

Unlike the unshielded case, the maximum combined stress is now well below the shakedown limit. The insertion of the thermal shield successfully reduced the heat generation within the vessel wall by absorbing a significant portion of the γ flux, the thermal gradient and the associated secondary stress have been reduced to an acceptable level. The vessel is therefore verified against both plastic collapse and thermal ratcheting/fatigue failure under Service Level A conditions.

VI. CONCLUSIONS

The analytical study conducted in this report confirms that the proposed structural design and thermo-mechanical configuration of the reactor pressure vessel and thermal shield fully comply with ASME Section III, Service Level A requirements. The initial sizing phase demonstrated that the vessel's wall thickness is primarily governed by external pressure stability criteria to prevent elastic collapse, leading to a final thickness of 15.22 cm.

It must be emphasized that this result represents the theoretical minimum thickness required to satisfy elastic shakedown and ratcheting criteria. In a practical design, this value must be increased to ensure an adequate safety margin and account for corrosion allowance, manufacturing tolerances, geometric imperfections, uncertainty in material properties and operating transient.

A critical finding of the analysis was the inadequacy of a non-shielded configuration; the vessel would have exceeded the elastic shakedown limit ($3S_m$) with a stress intensity of 510.18 MPa, potentially leading to failure via thermal ratcheting. The integration of a 2.13 cm thick thermal shield proved to be an effective mitigation strategy, successfully attenuating the γ flux and reducing the maximum combined stress intensity to 209.58 MPa, which is well within the safety margin of 372 MPa.

In conclusion, the design successfully integrates radiation shielding and structural integrity, guaranteeing the long-term reliability of the primary pressure boundary and the maintenance of the required cooling geometry under cyclic operational conditions.

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